6

Tools for Model Training and Experimenting

In this chapter, we will focus on creating training datasets and building baseline models. We start by building training sets from a training DataFrame and feature tables. You will learn how to combine feature table data with your training data without using traditional joins. We will also return to Databricks **AutoML** and explore how to use it to establish a baseline model quickly. We then cover how to experiment with different features, **hyperparameters**, and models when searching for predictive signals in your training data. Manually tracking configurations and their corresponding evaluation metrics is time-consuming. We introduce a component of MLflow called **MLflow Tracking**, which significantly improves tracking each permutation of parameters and the corresponding outputs.

We will highlight tools to integrate external data and models into your own projects and workflows, covering the Databricks Marketplace and the new AI Playground for readers interested in **Generative AI** (**GenAI**). In the *Applying our learning* section, we showcase how to put all the tools we have discussed to use in creating training datasets, building models, and enhancing projects across a variety of task types.

Here is what you will learn as part of this chapter:

- Building a training set from a feature table
- Baselining with AutoML
- Tracking experiments with MLflow
- Integrating external data and models
- Applying our learning